Cullen Kelley

703-732-0221 | Email | LinkedIn | Personal Site | GitHub

Professional Education

Masters of Applied IT, Cyber Security Concentration Aug. 2024 – Dec. 2025

George Mason University — GPA: TBD

Bachelor of Applied Science, Cyber Security Concentration Aug. 2021 – Aug. 2024

George Mason University — GPA: 3.83

- Phi Kappa Phi Honor Society Member, Dean's List (2022-2024)

Experience

Software Security Consultant Jun. 2024 – Aug. 2024

Affinity DB — Leesburg, VA

- Conducted quality assurance audit of numerous Python and VBA applications written by third party contractors prior to production integration
- Created application documentation and a writeup documenting the review process and sharing findings with stakeholders
- Audited using common industry tools including IDA CE, x64dbg, oletools, and DidierStevensSuite MS tooling

eCommerce Department Manager and Trainer Sep. 2020 – Jun. 2024

Safeway — Great Falls, VA

- Promoted from entry position eCommerce shopper within six months of hire due to exceptional performance and implemented improvements to department operations
- Identified vulnerabilities in multiple systems and responsibly disclosed them to Safeway IT security team
- Led operations of a three man team ensuring timely delivery of online orders with 99.3% on-timeness
- Launched eCommerce departments across Washington metropolitan area
- Independently produced training materials for on-boarding employees

Research & Technical Projects

Self-Driving Car BA Algorithm Research | *C++, Python, Google Ceres Solver* Sep. 2025 – Dec. 2025

- Assessed performance of bundle adjustment algorithm error rate and overhead using Oxford RobotCar
- Benchmarked Levenberg-Marquardt and Dogleg algorithms using sparse schur

Biometric Authentication Modalities Research | *IREX, FIPS 201-3, SP 800-76-2* Jan. 2025 – May 2025

- Conducted research on biometric authentication modalities including fingerprint, iris, and facial recognition
- Analyzed tradeoffs in performance, security, and usability of each modality

GhostRace CVE Research | *C, Ghidra, x64dbg* Jan. 2024 – May 2024

- Researched and developed a Proof-of-concept exploit for CVE-2024-2193 (GhostRace)
- Analyzed previous branch-misprediction and speculative execution CPU vulnerabilities Spectre and Meltdown

Machine Learning Malware Detector | *Python, Rust, MariaDB, Google Cloud* Aug. 2024 – Present

- Developed CI/CD pipeline for ingesting malware samples for supervised machine learning malware detection
- Ingested 201,549 Windows malware samples for training data using label encoder and padding each entry
- Achieved a malware detection accuracy of 94% with k-folded cross validation, comparing to train-test split baseline

Cloud Service Model Risk Assessment | *Google Cloud, AWS* Aug. 2024 – Dec. 2024

- Conducted research on induced security risks inherent to SaaS, PaaS, and IaaS service models
- Applied industry research in addition to NIST (SP) 800-210

Personal Cybersecurity Lab | *Suricata, Splunk, Docker, VMware Workstation* Feb. 2023 – Present

- Implemented a comprehensive SIEM (Security Information and Event Management) and SOAR (Security Orchestration, Automation, and Response) system for personal penetration testing projects
- Used for HackTheBox hacking labs and CTF events

Technical Skills / Accreditation / Awards

Certifications: Google Cybersecurity Professional Certificate, Security+ (exam scheduled for Jan.)
Languages: Rust, C, C++, GoLang, Python, PowerShell, SQL, VBA
Security Tools: OPNsense, Suricata, Wuzuh, Shuffle, Nessus, nMap, Ghidra, IDA CE, FireEye FLOSS, x64bdg, Autopsy, FTK Imager
Developer Tools: Git, Docker, Google Cloud Platform, Vim, Visual Studio
Additional Awards: Eagle Scout, Scouting Heroism Award, Sons of the American Revolution member